

# Finance Supervision 1

- 1) US Treasury Bills :  $\$50,000 \times 1.038^{20} = \$105,418.5588$   
Company Shares :  $\$50,000 \times 1.169^{20} = \$135,691.114$

Difference is  $\$1,030,272.56$

- 2)  $a \times 1.03^{10} = \$100,000$ . Solving for  $a$ :

$$a = \frac{\$100,000}{1.03^{10}} \quad a = 74,409.39149$$

You would need to put  $\$74,409.39$  into the savings account today in order to have  $\$100,000$  in 10 years

- 3) PV of annuity =  $C \left[ \frac{1}{r} - \frac{1}{r(1+r)^t} \right]$

$$\$10m \left[ \frac{1}{0.08} - \frac{1}{0.08(1.08)^{21}} \right] = \$100,168,031.6m$$

→ This is the value of the annuity excluding PV of sale price

$$\text{PV of Sale price} = \frac{\$20m}{1.08^{22}} = \$3.68m$$

Total PV is  $\$103,846,842$ . This is the Most you should invest in the mining operation in time 0.

- 4) All cash flows must be discounted using the cost of capital

$$NPV = \left[ \frac{\$250k}{1.13} + \frac{\$250k}{1.17} \right] - \left[ \frac{\$250k}{1.1} + \frac{\$250k}{1.12} \right]$$

$$NPV = \$783,220.41 - \$653,884.30$$

NPV is  $\$129,336 > 0$  So PGA should make the investment (provided there are no alternative mutually exclusive projects with a higher NPV).

$$5) \quad a) \quad NPV = \left[ \frac{\$500}{1.06} + \frac{\$1500}{1.06^2} + \frac{\$10,000}{1.06^{10}} \right] - \$10,000$$

$$NPV = \$7,390.64 - \$10,000$$

The NPV of this opportunity is  $-\$2,609.36$ .  $NPV < 0$  so you shouldn't accept

- (b) Apply the same equations, if the discount rate were 2% the NPV should be  $\$135.43$ , which you would accept as  $NPV > 0$ .

6.

NPV excluding environmental payments

$$-£10M + \frac{£3M}{1.05} + \frac{£3M}{1.05^2} + \dots + \frac{£3M}{1.05^8}$$

$$NPV = \cancel{£8,556,216.7} \quad £9,389,638.28$$

NPV accounting for perpetual payments

PV of ~~annuity~~ perpetuity :  $C_1 / r$

$$£12,000 / 0.05 = £240,000$$

$$NPV = £9,149,638.23$$

$NPV > 0$  so invest

7. Alphaville

$$NPV = \frac{\$2M}{0.08} - \$10M$$

$$NPV = \$15M$$

Betapod

$$NPV = \frac{\$1.5M}{1.08} + \frac{\$1.5M(1.02)}{0.08 - 0.02} - \$9M$$

$$NPV = \$1.38M + \$25.5M - \$9M$$

$$NPV = \$17.88$$

→ When  $r = 8\%$ , Betapod has a higher NPV, meaning it adds more wealth than Alphaville, so choose Betapod.

8.

(a)  $NPV = \frac{\cancel{€100}}{\cancel{1.095}} - \frac{€100}{0.095} - €1000$

$$NPV = €52.63 > 0 \text{ so he should invest}$$

(b)  $NPV = \frac{\cancel{€100}}{\cancel{0.095}} - \frac{€1000}{0.095} - €1000$        $€100/1.095 = €91.324$

If it takes a year to build, we need to discount the future cash flow before calculating perpetuity value

$$NPV = \frac{€91.32420091}{0.095} - €1000$$

$$NPV = \cancel{€38.56} \text{ so no we would recommend not investing.}$$

-€38.56